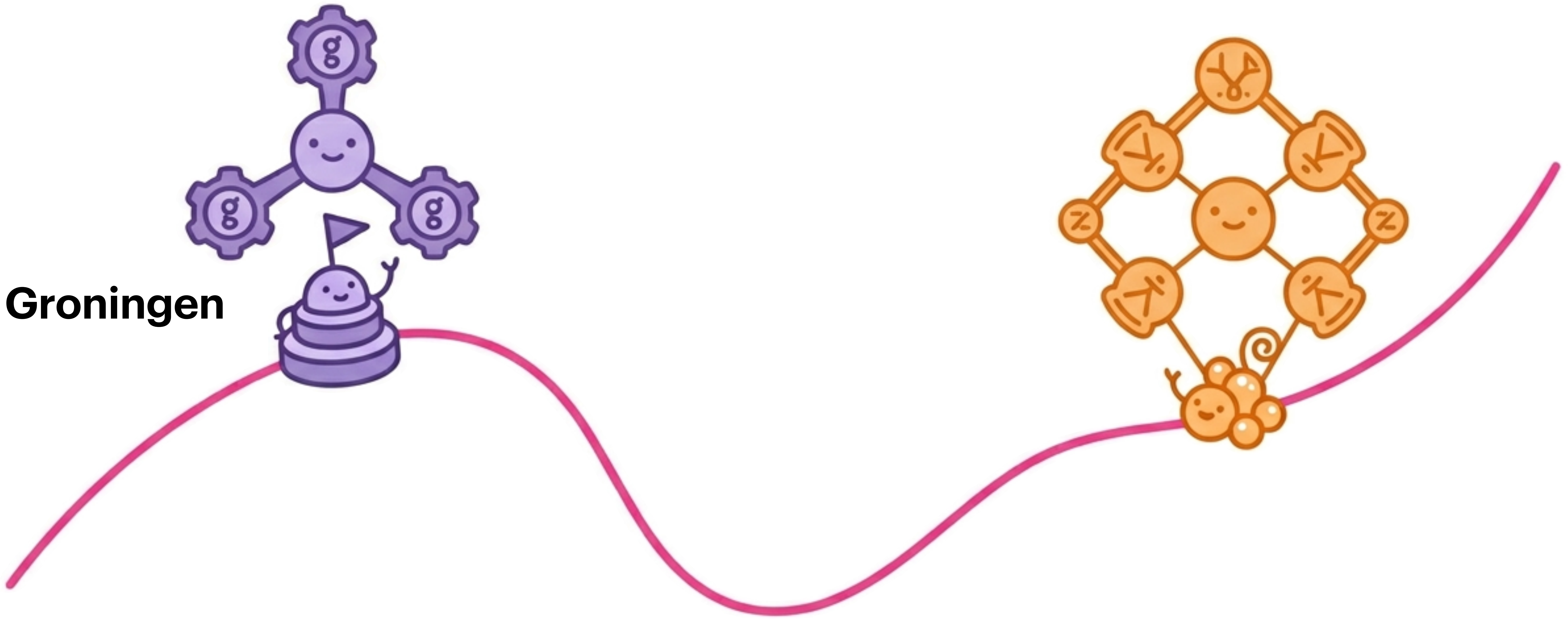
Stacky curves and the local-to-global principle



Juanita Duque-Rosero

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Joint work with Christopher Keyes, Andrew Kobin, Manami Roy, Soumya Sankar, and Yidi Wang.

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We can rescale to get infinitely many:

$$x = -3, y = -6, z = -3$$

$$x = 7,\ y = 14,\ z = 7$$

$$x = \lambda,\ y = 2\lambda,\ z = \lambda$$

Can you find nonzero integers x, y, and z such that

$$x^2 + y^2 = 5z^2 \, ?$$

Definition. A nonzero solution (x, y, z) is **primitive** if $\gcd(x, y, z) = 1$.

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NO: If such a solution existed, then there would be a solution to

$$x^2 + y^2 \equiv 0 \pmod 3.$$

$y \backslash x$	0	1	2
0	0	1	1
1	1	2	2
2	1	2	2

But then x and y are divisible by 3, which implies that z is too.

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Theorem [Hasse–Minkowski]. Let $f(x_1, \ldots, x_n)$ be a non-degenerate quadratic form, where the polynomial f has integer coefficients. Then $f(x_1, \ldots, x_n) = 0$ has a non-trivial solution over $\mathbb{Q}$ if and only if it has a nontrivial solution in $\mathbb{Q}_p$ for all primes p and one over $\mathbb{R}$.

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Exercise. Prove that for any $p > 7$, there is always a nonzero solution.

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Proof idea. Assume a solution exists, consider everything modulo 9, and prove that $3 \mid \gcd(x, y, z)$.

The difference:

$$x^2 + y^2 - 5z^2$$

$$x^2 + y^2 - 3z^2$$

Genus 0 curves

$$3x^3 + 4y^3 + 5z^3$$

Genus 1 curve

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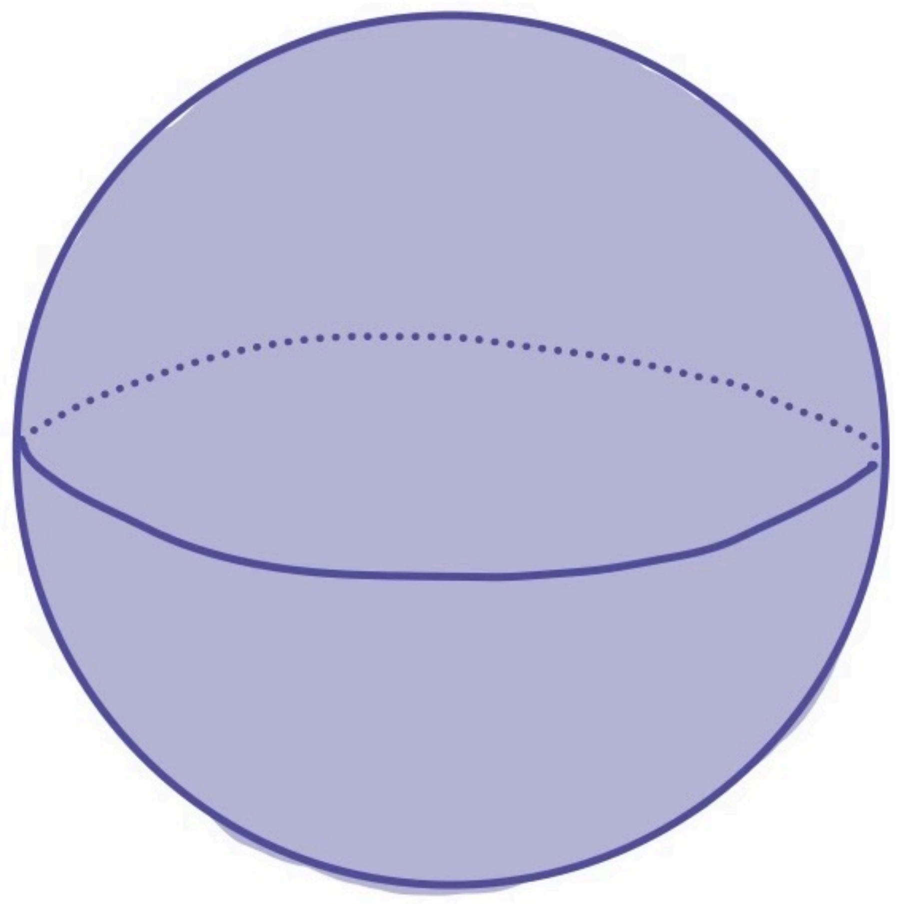

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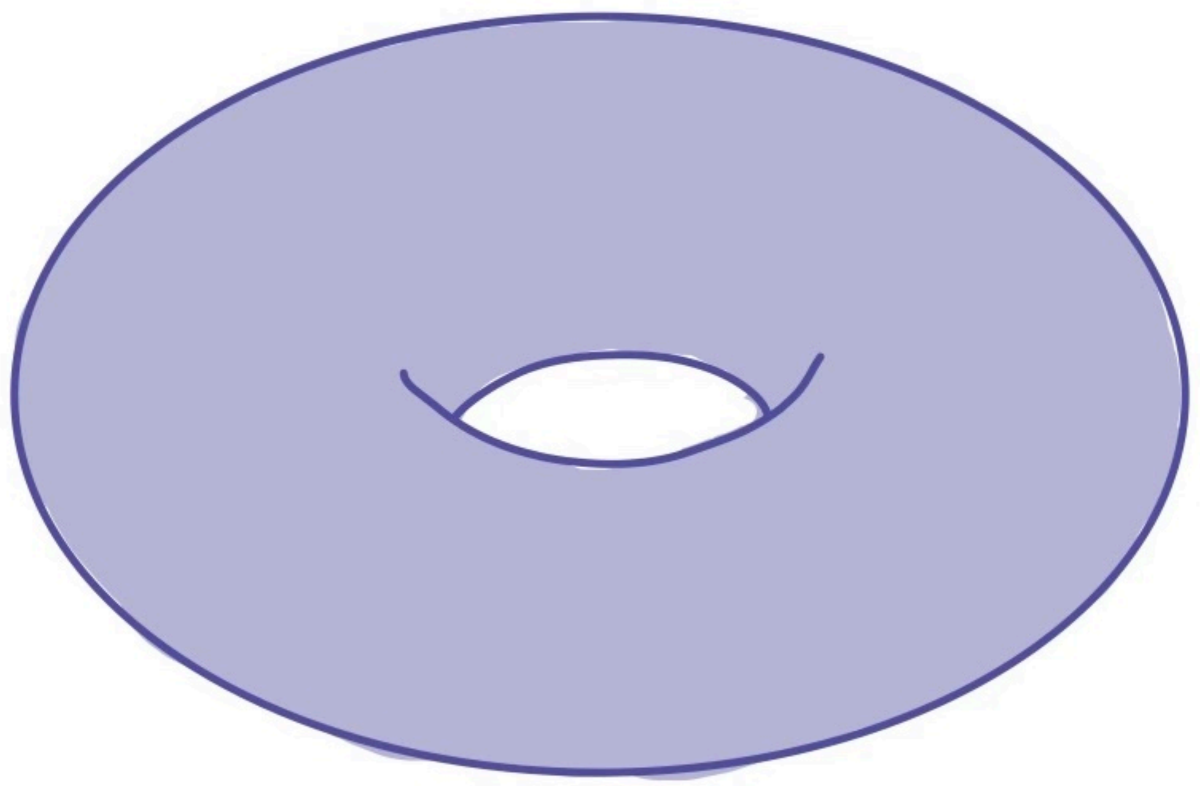

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A solution $(x_0, y_0, z_0) \in \mathbb{Q}^3$ to the equation $f(x, y, z) = 0$ is called a $\mathbb{Q}$-point on X.

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Note: We can also consider integral points (defined over $\mathbb{Z}$).

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Theorem [Hasse–Minkowski]. Let X be a smooth projective geometrically integral curve of genus 0 over $\mathbb{Q}$. Then X satisfies the local-to-global principle over $\mathbb{Q}$.

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Theorem [Bhargava '13]. A positive proportion of genus 1 curves in the weighted projective space given by $z^2 = f(x, y)$, where $f(x, y)$ is an integral binary quadratic form, violate the local-to-global principle over $\mathbb{Q}$.

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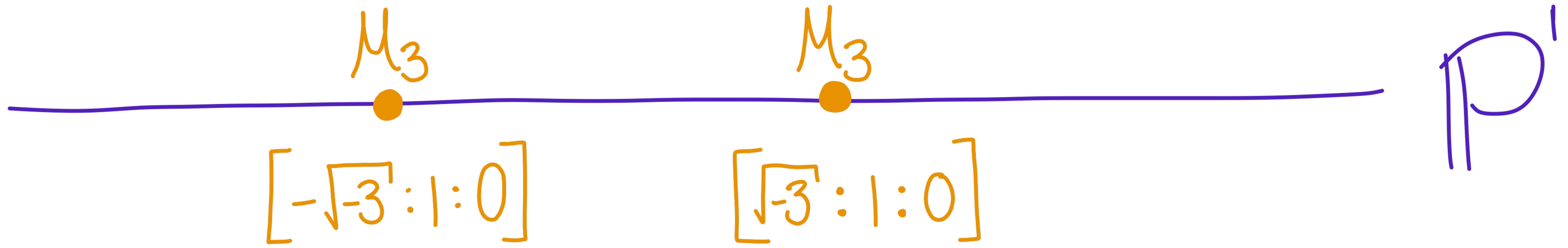

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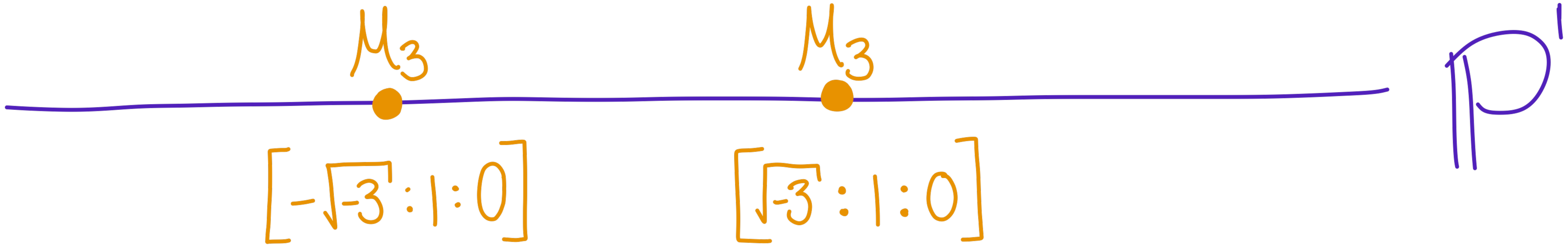

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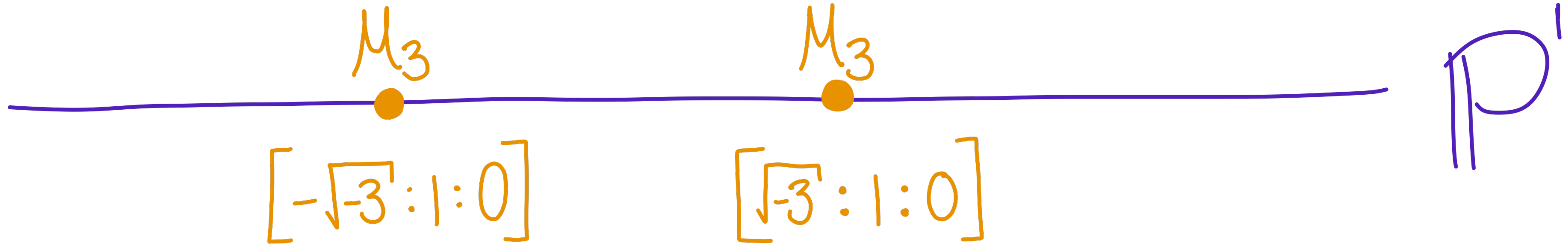
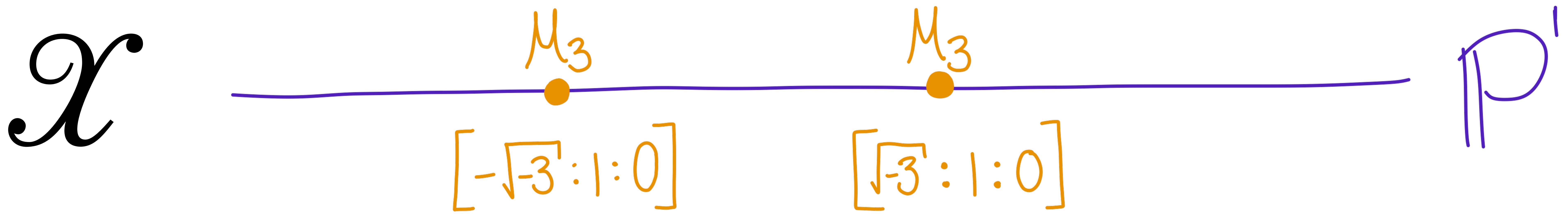


Primitive integral solutions to $f(x, y, z) = 0$ $\iff$ Integral points on $\mathscr{X}$: $\mathscr{X}(\mathbb{Z})$

Stacky curves

Definition. A stacky curve $\mathscr{X}$ over k is a smooth proper geometrically connected Deligne–Mumford stack of dimension 1 over k that contains a dense open subscheme.



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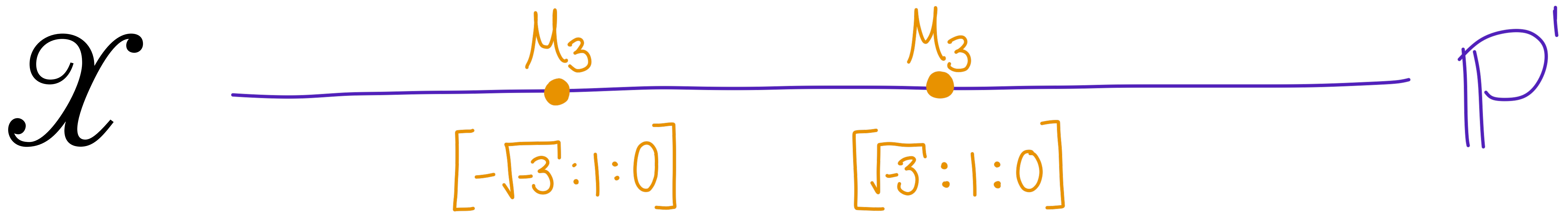

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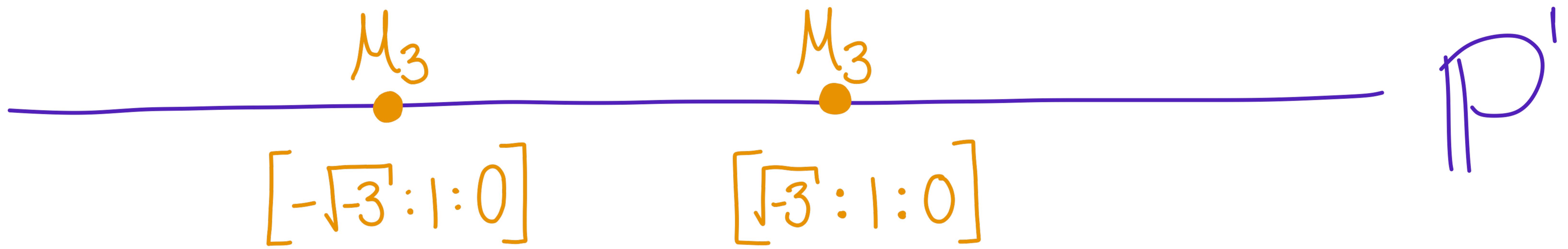

Claim: $\mathscr{X}$ is a stacky curve of genus $2/3$.

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Theorem [Bhargava & Poonen '21]. Let

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Consider the action of $\mathbb{G}_2$ as $[x : y : z] \mapsto [x : y : \lambda z]$ on $\mathscr{X} := [S/\mathbb{G}_m]$. Let $\mathscr{Y}$ be the quotient stack $[\mathscr{X}/\mathbb{G}_2]$. Then

(a) the genus of $\mathscr{Y}$ is $1/2$;

(b) $\mathscr{Y}(\mathbb{Z}_p) \neq \varnothing$ for every rational prime p and $\mathscr{Y}(\mathbb{R}) \neq \varnothing$;

(c) $\mathscr{Y}(\mathbb{Z}) \neq \varnothing$.

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Checking for local-to-global

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We need to answer:

1. $\mathscr{X}_{B,C,n}(\mathbb{Z}_p) \neq \emptyset$ for all primes p and $\mathscr{X}_{B,C,n}(\mathbb{R}) \neq \emptyset$?
2. $\mathscr{X}_{B,C,n}(\mathbb{Z}) \neq \emptyset$?

1. Are there local $\mathbb{Z}_p$- points and $\mathbb{R}$-points?

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Example [$x^2 + 29y^2 = 3z^3$].

The ideal 3 is split in $\mathbb{Q}\left(\sqrt{-29}\right)$. Thus there exist x and y such that

$$x^2 + 29y^2 \equiv 0 \pmod{3}.$$

Conclude $\mathcal{X}_{29,3,3}(\mathbb{Z}_3) \neq \varnothing$.

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(iii) $C\mathcal{O}_K = \mathfrak{j}\bar{\mathfrak{j}}\mathfrak{r}^2\mathfrak{i}^2$ with $\mathfrak{j}$, $\mathfrak{r}$, $\mathfrak{i}$ supported on split, ramified, and inert primes and $\left[\mathfrak{j} \cap \mathbb{Z}[f\sqrt{-B_0}]\right] \in \mathrm{Cl}\left(\mathbb{Z}[f\sqrt{-B_0}]\right)$.

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Brute force: $(9,0,3)$, $(10,10,10)$, …, but no primitive solutions.

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Computation of $R_K^\times/(R_K^\times)^3$ reveals no such d exist, so $\mathcal{X}_{29,3,3}(\mathbb{Z}) = \varnothing$.

How many curves $\mathscr{X}_{B,C,3}$ satisfy the local-to-global principle?

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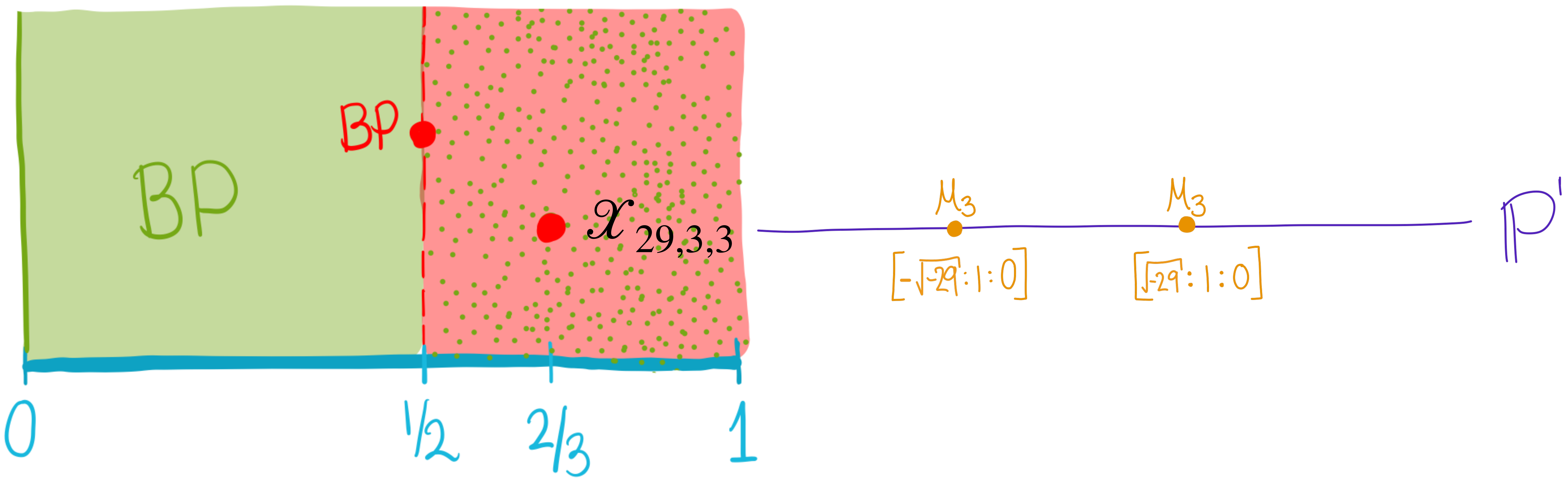
Theorem [DKKRSW '25]. There is a positive proportion of curves $\mathcal{X}_{B,C,3}$ that non-vacuously satisfy the local-to-global principle for integral points. That is,

$$\liminf_{T\to\infty} \frac{N(T)}{N^{\mathrm{loc}}(T)} > 0.$$

Thank you!



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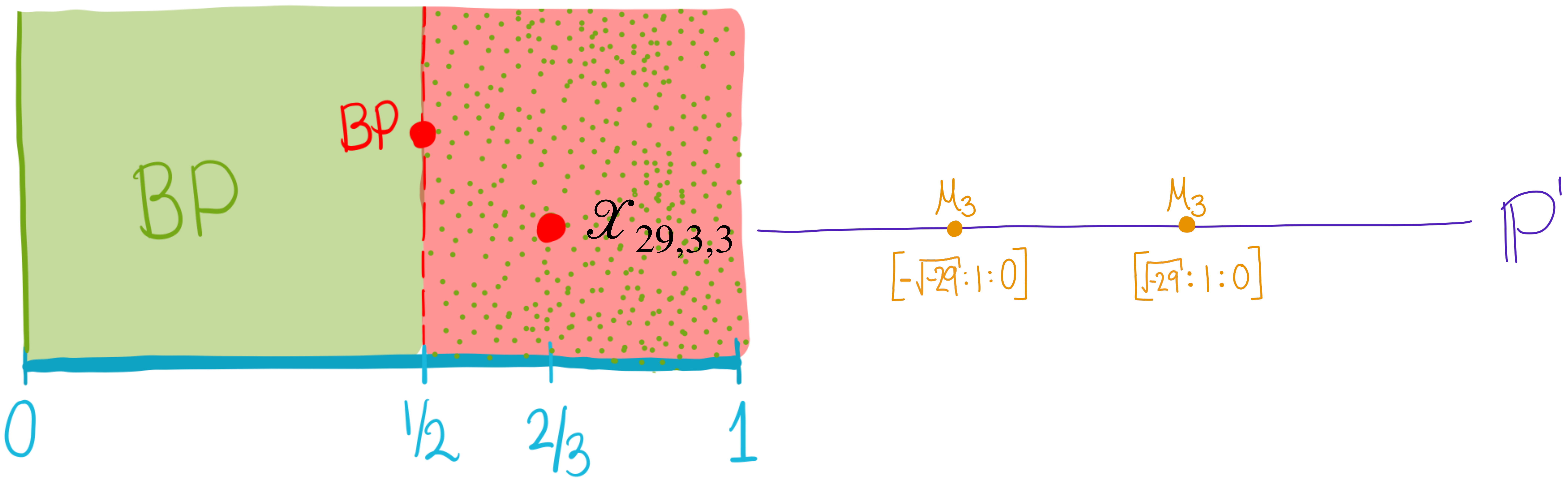

Theorem [DKKRSW '25]. There is a positive proportion of curves $\mathscr{X}_{B,C,3}$ that non-vacuously satisfy the local-to-global principle for integral points.

Q. OK, There are solutions... How many are there? Looking forward to Santi's talk tomorrow!